

SUNIT KUMAR

Senior Azure DevOps Engineer

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Profile

- 16+ years of IT experience, including 7+ years in Azure DevOps and cloud engineering and approximately 9 years in enterprise .NET application development.
- Strong expertise in designing and implementing CI/CD pipelines using Azure DevOps and GitHub Actions for automated, multi-environment application deployments.
- Experienced in CI/CD modernization, including migration from Jenkins to GitHub Actions and configuration of self-hosted Azure DevOps agents and GitHub Actions runners.
- Hands-on experience with Docker and Kubernetes, including application containerization, self-managed Kubernetes clusters, AKS, Kustomize, container registries, and workload deployments.
- Experienced in Infrastructure as Code using Terraform and ARM Templates, including modular provisioning, remote state management, and environment-specific configurations.
- Strong hands-on experience with Azure PaaS, identity, database, messaging, monitoring, and container services.
- Strong automation experience using PowerShell, PowerShell Remoting, and Microsoft Graph API for deployments, IIS operations, Azure SQL migration, Entra ID/B2B administration, and access governance.
- Implemented secure DevOps practices using Workload Identity Federation, HashiCorp Vault, AppRole authentication, Kubernetes Secrets, NGINX Ingress, and TLS/HTTPS.
- Implemented monitoring and centralized logging using Prometheus, Grafana, Fluent Bit, Elasticsearch, Kibana, and Application Insights for cloud and Kubernetes workloads.
- Delivered significant automation and modernization initiatives, including .NET application upgrades and automation of a ~250 GB Azure SQL database migration across Microsoft and Shell subscriptions using PowerShell, reducing 15+ hours of manual effort.

Education

- Master of Computer Application (MCA) from Vellore Institute of Technology, Vellore, 2005-2008

Technical Expertise

Primary Expertise (Azure DevOps, Azure Cloud, Cloud-Native Engineering):

Azure DevOps, GitHub Actions, Terraform, Docker, Kubernetes, AKS, PowerShell, ARM Templates, Kustomize, HashiCorp Vault

Azure Services

Azure App Service, Azure Functions, ACR, Azure Storage, Azure SQL, Azure Service Bus, Azure Key Vault, Microsoft Entra ID, Application Insights

Observability: Prometheus, Grafana, Fluent Bit, Elasticsearch, Kibana

Secondary Skills (.NET Full Stack Development):

C#, .NET Framework, .NET Core/.NET 6, ASP.NET MVC/Web API, SQL Server, Entity Framework

Employment History

- Independent DevOps Engineering | Oct 2025 – Present
- Encora Innovation Labs | Senior Azure DevOps Engineer | Dec 2023 – Sep 2025 (GoDaddy Client)
- T-Systems ICT India Pvt. Ltd. | Azure DevOps Engineer | Dec 2018 – May 2023 (Royal Dutch Shell - SBO Bangalore)
- NTT Data Services | Senior Software Engineer | Nov 2016 – Mar 2018
- Johnson Controls Pvt. Ltd. | Senior Software Engineer | Jan 2014 – Oct 2016
- Standard Chartered Scope International Pvt. Ltd. | Product Analyst | Sep 2011 – Dec 2013

- Pellucid Healthcare Networks Pvt. Ltd. | Software Engineer | Nov 2008 – Aug 2011

Projects

➤ **My Health Clinic – Self-Hosted Kubernetes Platform | Jan 2026 – Present | Live: (<https://mhc.sunitdevops.site>)**

Description:

MHC is a cloud-native healthcare application deployed on a self-hosted Kubernetes platform with automated CI/CD, Infrastructure as Code, secrets management, observability, and multi-environment delivery.

Key Responsibilities & Individual Contributions:

- Migrated the My Health Clinic application to .NET 6 and upgraded the Angular frontend, resolving migration and compatibility issues.
- Decoupled MyHealth.Web and MyHealth.API by replacing direct assembly references with REST API communication and implementing JWT authentication.
- Developed and integrated user registration functionality between the Angular frontend and MyHealth.API.
- Built a 4-node self-hosted Kubernetes cluster with one control-plane and three worker nodes using kubeadm, containerd, and Calico.
- Containerized MyHealth.Web, MyHealth.API, and DACPAC database deployment using Docker and published versioned images to Azure Container Registry (ACR).
- Managed Dev, Pre-Prod, and Prod Kubernetes deployments using namespaces and Kustomize base/overlay configurations.
- Designed a multi-stage Azure DevOps YAML pipeline to automate Terraform validation, Docker builds, image publishing, and Kubernetes deployments across Dev, Pre-Prod and Prod.
- Implemented Terraform with an Azure Storage remote backend and environment-specific state management.
- Integrated HashiCorp Vault AppRole authentication with Azure DevOps to securely populate environment-specific Kubernetes Secrets.
- Automated SQL database deployment using a containerized DACPAC Kubernetes Job as part of the CI/CD workflow.
- Configured secure public access using Cloudflare DNS and Cloudflare Tunnel, NGINX Ingress, cert-manager, and Let's Encrypt TLS certificates.
- Implemented monitoring with Prometheus and Grafana and centralized logging with Fluent Bit, Elasticsearch, and Kibana.
- Configured Azure DevOps service connections using Workload Identity Federation for secure access to Azure resources without maintaining long-lived client secrets.
- Installed and configured a **self-hosted Azure DevOps Agent and agent pool** to execute build, Terraform, Docker, and Kubernetes deployment jobs for the MHC CI/CD pipeline.

Tech Stack: Azure DevOps, GitHub, Terraform, Docker, Kubernetes, Kustomize, Azure Container Registry (ACR), HashiCorp Vault, NGINX Ingress, Prometheus, Grafana, Fluent Bit, Elasticsearch, Kibana, Cloudflare Tunnel, .NET 6, Angular, SQL Server

➤ **Employee Recruitment Portal | Oct 2025 – Dec 2025 | Live: (<https://erp.sunitdevops.site>)**

Description:

Employee Recruitment Portal is a .NET-based web application for managing job openings, candidate referrals, CV screening, interviews, and job offers. The project demonstrates automated CI/CD deployment to a traditional Windows/IIS environment.

Key Responsibilities & Individual Contributions:

- Designed and implemented an Azure DevOps YAML CI/CD pipeline to build, publish, and deploy the ERP application to IIS on a Windows 11 host.
- Used a self-hosted Azure DevOps Agent to execute application and database deployment jobs.
- Configured and managed the IIS website, application pool, deployment directory, and application bindings using PowerShell.

- Implemented artifact-based application deployment and automated SQL database deployment using DACPAC and SqlPackage.
- Used an Azure DevOps Library Variable Group (ERP-DEV) to manage DEV-specific database, deployment-path, IIS, and configuration values.
- Used PowerShell Remoting for application backup, file transfer, configuration updates, IIS restart, and post-deployment validation.
- Configured public access through Cloudflare DNS and Cloudflare Tunnel for the live application at erp.sunitdevops.site.
- Redesigned the Login and Home pages and resolved UI and grid-alignment issues to improve application usability and presentation.

Tech Stack: Azure DevOps, GitHub, PowerShell, PowerShell Remoting, Self-Hosted Agent, IIS, SQL Server, DACPAC/SqlPackage, Cloudflare Tunnel, .NET

➤ **Encora Innovation Labs | Senior Azure DevOps Engineer | March 2025 – Sep 2025 (GoDaddy Client)**

Ecom Message Processor

Description:

Ecom Message Processor is a Windows service deployed via Chocolatey that retrieves entitlement data and integrates with GDAPI to create or update entitlement records.

Key Responsibilities & Contributions:

- Migrated the Ecom Message Processor deployment from Jenkins to GitHub Actions, converting the existing Jenkinsfile into an automated GitHub Actions workflow.
- Integrated PowerShell and Chocolatey commands to automate Windows service installation and upgrades.
- Used PowerShell Remoting to connect to target servers and deploy and update the Ecom Message Processor service.
- Configured environment-specific deployment workflows for Dev, Test, and Prod, with secure management of GitHub secrets.
- Automated the end-to-end deployment process, reducing manual intervention and improving release consistency.

Tech Stack: Jenkins, GitHub Actions, PowerShell, PowerShell Remoting, Chocolatey, Windows Services, .NET Core

➤ **Encora Innovation Labs | Senior Azure DevOps Engineer | Dec 2023 – Feb 2025 (GoDaddy Client)**

GDAPI Migration & CI/CD Modernization (GoDaddy API Integration)

Description:

GDAPI is a GoDaddy API platform used to automate hosting and domain-management operations including DNS, SSL, FTP, database, and account-management activities.

Key Responsibilities & Contributions:

- Migrated the legacy GDAPI application from .NET Framework to .NET Core using .NET Upgrade Assistant.
- Replaced the existing Jenkins pipeline with GitHub Actions and implemented an automated CI/CD workflow.
- Containerized the application using Docker to provide consistent builds and deployments across environments.
- Set up and configured a self-managed Kubernetes cluster for the Development zone and deployed containerized GDAPI services.
- Validated deployed APIs, application integrations, and Kubernetes workloads in the Development environment.
- Configured Development, Testing, and Production environments in GitHub Actions to support controlled releases.
- Installed and configured GitHub Actions runners to execute build and deployment jobs in an isolated environment.

- Structured CI/CD workflows into separate build, test, and deployment stages for improved maintainability and failure isolation.
- Integrated existing PowerShell automation scripts into GitHub Actions for environment setup and deployment.

Tech Stack: GitHub Actions, Jenkins, Docker, Kubernetes, PowerShell, ASP.NET Core Web API, .NET Core, SQL Server

➤ **T-Systems ICT India Pvt. Ltd. | Azure DevOps Engineer | Dec 2018 – May 2023 (Royal Dutch Shell Client)**
Shell Lube Analyst

Description:

Enterprise oil condition monitoring platform supporting end-to-end sample workflow processing, reporting, and master data management for internal teams and external customers.

Key Responsibilities & Contributions:

- Designed and implemented Azure DevOps CI/CD pipelines to automate build and release processes across multiple environments.
- Independently provisioned and configured the Staging environment in Azure to support testing and controlled deployments.
- Provisioned Azure PaaS services including Azure App Services, Azure Functions, Storage, Service Bus, Azure SQL, and Logic Apps using ARM Templates.
- Automated a ~250 GB Azure SQL database migration across Microsoft and Shell subscriptions using PowerShell, reducing 15+ hours of manual effort.
- Administered Microsoft Entra ID B2B collaboration and resolved 500+ external-user authentication and access issues.
- Managed production deployments and hyper-care activities, ensuring release stability and minimal downtime.
- Implemented Point-to-Site (P2S) VPN connectivity for secure access to Azure VMs.
- Monitored Azure workloads using Application Insights and Azure monitoring capabilities and resolved cloud/deployment-related production issues.
- Migrated infrastructure provisioning from ARM Templates to modular Terraform, improving maintainability and scalability.
- Worked with Azure Kubernetes Service (AKS) for containerized workload deployment and monitoring in coordination with platform teams.
- Developed PowerShell automation using Microsoft Graph API for Entra ID user management, B2B collaboration, and access-governance activities.

Tech Stack: Azure DevOps, Azure Repos, Terraform, ARM Templates, AKS, Azure App Service, Azure Functions, Azure SQL, Azure Storage, Azure Service Bus, Azure Key Vault, Microsoft Entra ID, Application Insights, PowerShell, Microsoft Graph API

Earlier .NET Development Experience | 2008–2018

Before transitioning to Azure DevOps and cloud engineering, gained approximately **nine years of enterprise .NET development experience** across healthcare, banking, engineering, and financial services domains.

- Developed web applications, Windows services, APIs, database solutions, and integration workflows using C#, ASP.NET MVC, .NET Framework, SQL Server, Entity Framework, SOAP Web Services, TFS, and JavaScript.
- Modernized BMW Financial Services integrations by replacing BizTalk workflows with .NET services and queue-based processing.
- Designed and developed the Johnson Controls Engineering Change Notification application, including database architecture, workflow modules, and legacy-data migration.
- Delivered enterprise application development and production support for Standard Chartered and Pellucid Healthcare.

Recognition: Quarterly awards for the Shell Lube Analyst project – Dec 2019 and Mar 2021.
